

ClimRisk

Climate Risk Intelligence Engine Methodology Overview — Physical, Transition and Financial Risk Layers

Version 0.2 | May 2026 | Confidential

Prepared for: Partners

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Document scope

This document describes the methodology underlying the ClimRisk CRI engine. Section 1 sets out the three-pillar architecture. Sections 2 to 4 cover physical climate risk in full technical depth, which is the primary scope of this review. Sections 5 and 6 provide structured orientation to the transition risk and financial risk layers, describing how each is constructed and how the two integrate with the physical pillar to produce dollar-denominated, scenario-conditioned output.

Seismic and earthquake risk is explicitly excluded from climate hazard outputs per TCFD and IFRS S2 definitions. All hazard scope, data sources, and known limitations are documented within.

1. Three-Pillar Architecture

The CRI engine runs a single integrated pipeline across three sequential pillars. Each pillar is independently auditable and produces its own scored output; together they feed a composite CRI rating and a climate-adjusted financial valuation.

Pillar	Weight	Primary outputs	Key data sources
1. Physical Climate Risk	35%	Per-hazard severity (0 to 5), annual production loss %, physical risk score (0 to 100)	WRI Aqueduct 4.0; Open-Meteo CMIP6 MRI-AGCM3.2-S; NASA POWER MERRA-2; IPCC AR6
2. Transition Risk	35%	EBITDA compression under each SSP; carbon cost trajectory; stranded-asset ratio; break-even carbon price	NGFS Phase 4; IEA WEO / NZE 2050; CDP emissions disclosure
3. Financial Risk	30%	Climate-adjusted NPV; implied share-price delta vs market; FCF trajectory; scenario divergence fan chart	Company financials; WACC model; climate risk premium beta-calibration

The composite CRI rating (0 to 100, mapped to an AAA–D letter scale) is the weighted mean of the three pillar scores. It is designed to be directly comparable with credit ratings and to satisfy the scenario-analysis requirements of TCFD, IFRS S2, CSRD, and ECB/EBA Pillar 3 climate disclosures.

2. Physical Climate Risk — Scope and Hazard Taxonomy

The physical risk pillar assesses ten climate-driven hazards per asset location across four IPCC-aligned SSP scenarios and any chosen horizon year. Seismic and earthquake risk is explicitly excluded: it is a geophysical (tectonic) hazard that is not climate-driven, and its inclusion would be flagged as a scope error under TCFD and IFRS S2 definitions. Seismic data is computed separately and surfaced as an informational footnote for enterprise risk management purposes only.

2.1 Ten-hazard taxonomy

No.	Hazard	Type	Primary driver	IPCC and regulatory basis
1	Heat Stress	Acute / Chronic	WBGT index; extreme heat days above 35 deg C wet-bulb	IPCC AR6 WG2 Ch2, Ch9; TCFD acute physical
2	Riverine Flood	Acute	Heavy precipitation and river overflow; return period analysis	WRI Aqueduct 4.0; IPCC AR6 WG2 Ch11
3	Coastal Flood	Acute	Storm surge plus sea level rise compound event	IPCC AR6 WG1 Ch9; IPCC AR6 WG2 Ch3
4	Sea Level Rise	Chronic	GMSL projections; land subsidence; chronic inundation	IPCC AR6 WG1 Table 9.9; NGFS
5	Saltwater Intrusion	Chronic	Coastal aquifer salinisation; SLR and tidal encroachment	FAO AQUASTAT; IPCC AR6 WG2 Ch3
6	Landslide	Acute	Slope instability under extreme precipitation; terrain proxy	IPCC AR6 WG2; Copernicus DEM 30m
7	Wildfire	Acute	Fire weather index (FWI); vegetation fuel load by LULC type	NASA FIRMS; IPCC AR6 WG2 Ch10
8	Cyclone / Tropical Storm	Acute	Wind intensity and storm surge; latitude-gated 5 to 25 deg N/S	IBTRACS; IPCC AR6 Ch11
9	Drought	Chronic / Acute	Meteorological and hydrological drought; PDSI analogue	WRI Aqueduct 4.0; IPCC AR6 WG2
10	Water Stress	Chronic	Freshwater withdrawal vs availability; operational dependency	WRI Aqueduct 4.0 (Kuzma et al. 2023)

Each hazard returns three values: a severity index on the WRI-aligned 0 to 5 scale; an annual probability of a damaging event; and an expected annual production loss percentage. Hazards are marked not applicable where site physics eliminate the exposure. Coastal flood and sea level rise are suppressed for inland assets more than 80 km from the coast or above 20 m elevation. Landslide is suppressed on flat terrain.

3. Spatial Downscaling — 25 km Grid to 1 km Asset Precision

Most physical risk tools snap assets to national or regional averages. The CRI engine queries climate data at the asset's actual coordinates and applies sub-grid terrain corrections to achieve 1 km precision. The method is a five-layer pipeline.

3.1 Layer 1 — WRI Aqueduct 4.0 regional baseline

WRI Aqueduct 4.0 (Kuzma et al. 2023) provides the foundational hazard index for water stress, riverine flood, coastal flood, and drought across more than 80 country and region codes. Indices are blended from the nearest available regional centroids using inverse-distance-squared weighting, capped at 2,000 km. This cap prevents climatologically unrepresentative regions from contaminating the baseline, a failure mode present in many vendor implementations.

3.2 Layer 2 — NASA POWER MERRA-2 observed baseline

The NASA POWER API (MERRA-2 reanalysis, 2001 to 2020 climatological mean, 0.5 degree or approximately 55 km resolution) provides the observed baseline at the asset's actual grid cell: mean and maximum 2-m air temperature, daily precipitation, and relative humidity. This replaces static look-up tables with coordinate-specific observations.

3.3 Layer 3 — CMIP6 model-consistent delta via Open-Meteo

Climate warming is computed using a delta-downscaling approach that avoids the systematic GCM mean-state bias. Comparing raw CMIP6 model output directly to observed reanalysis produces offsets of 10 to 20 degrees C in tropical regions due to differences in grid resolution and sampling. Instead, the engine computes a model-consistent delta from the same model and the same 0.25 degree grid cell:

```
warming_delta = T2M_MAX(CMIP6, future year) - T2M_MAX(CMIP6, historical ref. 2010)
[ same model, same 0.25 deg grid cell ]

projected_T2M = observed_baseline (NASA POWER) + warming_delta x SSP_scale_factor
```

The CMIP6 source is the Open-Meteo Climate Change API using the MRI-AGCM3.2-S model (Mizuta et al. 2017, Journal of the Meteorological Society of Japan) at 0.25 degree or approximately 25 km native resolution. This is the same CMIP6 dataset family as NASA NEX-GDDP-CMIP6, accessed live at exact asset coordinates rather than pre-gridded.

3.4 SSP scenario scaling — IPCC AR6 WG1 Table 4.5

The Open-Meteo CMIP6 API provides projections natively under SSP5-8.5. Warming deltas for other SSPs are derived by scaling the raw delta using GMST ratios from IPCC AR6 Working Group 1 Table 4.5, with a time-interpolated blend from 2026 (scenarios converged) to 2050 (full scenario divergence):

SSP scenario	IPCC GMST at 2050	Scale vs SSP5-8.5	NGFS alignment
SSP5-8.5	2.00 deg C	1.000 (baseline)	Hot House / stress test
SSP3-7.0	1.70 deg C	0.850	Current Policies
SSP2-4.5	1.35 deg C	0.675	Delayed Transition

SSP scenario	IPCC GMST at 2050	Scale vs SSP5-8.5	NGFS alignment
SSP1-2.6	0.85 deg C	0.425	NZE 2050 / Net-Zero

```

year_weight = (target_year - 2026) / (2050 - 2026) [ clamped 0 to 1 ]
SSP_scale = 1.0 - year_weight x (1.0 - SSP_scale_at_2050)
warming_c = max(0.0, raw_delta x SSP_scale)

```

3.5 Layer 4 — Sub-grid elevation correction

Elevation data (Copernicus DEM 30 m proxy) is used to apply an environmental lapse rate correction of 6.5 deg C per 1,000 m, adjusting the CMIP6 grid-cell temperature to the asset's actual terrain altitude. This is particularly material for highland mining operations, high-altitude agricultural sites, and Andean or Alpine assets where the grid-cell mean elevation differs substantially from the facility location.

3.6 Layer 5 — Site physics modifiers

Four site-physics corrections are applied at 1 km resolution to produce the final hazard scores:

Modifier	Driver	Effect on hazard scores
Coastal proximity	Haversine distance to coastline; threshold below 80 km	Activates coastal flood, sea level rise, saltwater intrusion, and cyclone hazards; applies sea-breeze temperature moderation
Urban heat island	NASA MODIS / HLS LULC proxy, urban classification	Adds 1.5 to 2.5 deg C to local heat stress baseline; amplifies extreme heat day frequency
Slope and terrain	SRTM DEM gradient proxy; elevation change per unit distance	Scales landslide probability; slope factor above 1.2 triggers elevated landslide exposure
River proximity	Haversine distance to major river systems	Scales riverine flood amplitude; proximity multiplier of 1.4 at high proximity scores

4. Joint Probability Model and Scoring

4.1 Annual production loss

Each applicable hazard h produces an expected annual production loss probability $p(h)$. The joint annual production loss across all applicable hazards uses the standard actuarial independence-adjusted formula:

$$\text{annual_production_loss} = 1 - \text{product over all } h [1 - p(h)]$$

This correctly handles co-occurrence: two hazards each with a 5% annual loss probability produce a 9.75% combined loss, not 10% as a naive sum would give. For assets where multiple high-severity hazards co-occur, for example a coastal refinery facing both tropical cyclone and heat stress, the compounding effect is material.

4.2 Sector-specific sensitivity adjustments

Hazard severity scores are adjusted by sector-specific sensitivity multipliers before entering the joint-loss calculation. The multipliers reflect differential operational exposure: a semiconductor fab has extreme sensitivity to water stress; an open-pit mine has high landslide and heat exposure but minimal coastal flood sensitivity; a data centre has elevated heat stress and cooling-water sensitivity. All multipliers are documented in versioned configuration files, not embedded in code.

4.3 Physical risk score (0 to 100)

$$\text{physical_risk_score} = \text{mean}(\text{severity}(h) \text{ for all applicable } h) \times 20$$

in range [0, 100]

This maps the 0 to 5 WRI-aligned severity scale directly to a 0 to 100 score. Scores at or above 55 are flagged high risk; at or above 35 are elevated; below 35 are moderate to low. The score is consistent across the Supply Chain, Portfolio, and Climate Rating panels to prevent the internal discrepancies common in multi-source risk platforms.

4.4 Precipitation delta — hyperarid guard

When computing percentage changes in precipitation for drought and flood hazards, two guards are applied to prevent model artefacts from propagating into scores:

- Minimum denominator threshold: if CMIP6 historical precipitation is below 0.10 mm per day (approximately 36 mm per year), precipitation percentage change is suppressed and set to null. This prevents extreme percentage amplification at hyperarid sites such as the Atacama or Sahara, where a small absolute change translates to an astronomically large percentage.
- Change cap: raw precipitation delta is clamped to the range of negative 90% to positive 200% to exclude CMIP6 model artefacts at the tails of the distribution.

Source: Open-Meteo CMIP6 MRI-AGCM3.2-S; Mizuta et al. 2017 J. Meteor. Soc. Japan; IPCC AR6 WG1 Table 4.5; WRI Aqueduct 4.0 Kuzma et al. 2023; NASA POWER MERRA-2; IBTRACS; NASA FIRMS; Copernicus DEM 30m.

4.5 Physical risk output summary

Output	Definition	Unit
Hazard severity	Per-hazard WRI-aligned index	0 to 5
Annual production loss	Joint actuarial loss across applicable hazards	Percent of annual production
Physical risk score	Mean severity multiplied by 20	0 to 100
Warming delta	T2M_MAX future minus T2M_MAX historical, same CMIP6 model	deg C
Precipitation delta	Percent change in daily mean precipitation, CMIP6 model-consistent	Percent; null if hyperarid
Top hazards	Hazards ranked by severity contribution	Ordered list, top 3
Downscaling trace	Full audit trail: data sources, grid cell, SSP scale factors	Structured log

5. Transition Risk Layer — Construction Overview

Scope note: This review focuses on physical climate risk (Sections 2 to 4). This section describes the transition risk layer for orientation, covering how it is constructed, what it models, and how it connects to the financial pillar. Full technical documentation is available on request.

5.1 Scenario set

The transition layer runs four NGFS Phase 4 and IPCC-aligned scenarios:

Scenario	SSP	Carbon price by 2050	Narrative
NZE 2050	SSP1-2.6	approx USD 250 per tCO ₂ e	Aggressive, front-loaded; green demand tailwinds for critical minerals
Below 2 deg Orderly	SSP2-4.5	approx USD 160 per tCO ₂ e	Smooth glide path; reference case for most financial models
Delayed Transition	SSP3-7.0	approx USD 100 per tCO ₂ e then abrupt repricing ca 2030	Policy lag then abrupt repricing; highest stranded-asset risk
Current Policies	SSP5-8.5	below USD 50 per tCO ₂ e formally	Hot-house baseline; physical risk dominates over transition costs

5.2 Carbon cost mechanism

For each scenario and year, the company's carbon cost is:

```
carbon_cost[company, year]
= scope1_2_emissions[company, year]
x carbon_price[scenario, year]
x carbon_price_coverage[company, region, year] in range [0, 1]
x (1 - free_allocation[company, region, year])
```

Coverage and free allocation are region- and sector-specific versioned parameters reflecting ETS design differences across the EU, UK, Australia, and emerging schemes. Scope 3 emissions are tracked and routed through the supply-chain demand path rather than directly into cost, consistent with GHG Protocol scope distinctions.

5.3 Commodity demand and price shocks

Revenue is decomposed by commodity. For each commodity and scenario the engine applies four adjustments:

- Demand elasticity: volume adjusts via demand_elasticity[k] multiplied by the change in demand index from IEA and NGFS scenario databases.
- Green premium or brown discount: scenario-specific price adjustments for transition commodities such as copper, lithium, and aluminium versus fossil commodities such as coal and oil.
- Physical disruption: volume is further reduced by the annual production loss from the physical risk pillar, creating a direct linkage between the two layers.
- Asset-level tipping points: when a mine's unit cost exceeds the scenario commodity price, utilisation drops to zero and a stranded asset write-down is triggered. This non-linearity is critical for high-cost resource assets under aggressive transition scenarios.

5.4 Marginal abatement cost curves (MACC)

Transition capex is modelled against the company's marginal abatement cost curve, which represents the cost per tonne of CO₂ abated for each available decarbonisation lever including fuel switching, energy efficiency, electrification, and CCS. The MACC is a first-class auditable input exposed directly to users. This is the most defensible and differentiated element of the transition pillar: it makes abatement cost explicit and traceable rather than burying it in a composite score.

6. Financial Risk Layer — Construction Overview

Scope note: This section provides structural orientation to the financial layer, describing the income statement, FCF, WACC adjustment, and DCF methodology at a level appropriate for a technical review of the physical risk pillar. Full financial pillar documentation is available on request.

6.1 EBITDA bridge — both risk layers combined

Both physical and transition risks enter the income statement explicitly:

```
EBITDA[company, year]
= revenue[company, year]
- opex[company, year]
- carbon_cost[company, year]
- physical_loss_cost[company, year]

opex breakdown:
energy_cost + labour + materials
+ water_cost[year] [ scaled by WRI water-stress index ]
+ logistics[year] [ disrupted by extreme weather events ]

physical_loss_cost = annual_production_loss x unit_margin
```

This decomposition makes the climate channel of each cost line explicit and auditable. Investment committees can interrogate which cost line drives the EBITDA compression under each scenario.

6.2 Free cash flow

```
FCF[company, year]
= NOPAT[company, year] + DA[company, year]
- maintenance_capex
- transition_capex[company, year] [ MACC-based abatement spend ]
- adaptation_capex[company, year] [ hazard-driven physical adaptation ]
- change_in_working_capital[company, year]

DA[company, year]
= baseline_DA
+ DA on transition_capex and adaptation_capex
+ accelerated_DA on stranded assets
```

Adaptation capex crowding out growth capex is an explicitly modelled non-linearity: forced adaptation spend displaces expansion capex, reducing long-run volume growth. This is material for capital-intensive miners and energy companies in high-physical-risk locations.

6.3 Climate-adjusted WACC

```
WACC_adj[company, scenario]
= WACC_base[company] + climate_risk_premium[company, scenario]

climate_risk_premium
= beta_transition x transition_exposure[company]
+ beta_physical x physical_exposure[company]
+ beta_policy x regulatory_uncertainty[company, scenario]
```

Beta coefficients are calibrated empirically where CDS spreads and bond yield data exist for comparable climate exposures, and via documented judgement where they do not. All beta values are versioned and auditable. The WACC adjustment is the mechanism by which transition and physical risk premia propagate into the discount rate, compounding their effect on long-horizon NPV.

6.4 Climate-adjusted DCF and outputs

```
NPV[company, scenario]
= sum over t [ FCF[company, t] / (1 + WACC_adj)^(t - T0) ]
+ TerminalValue[company, scenario] / (1 + WACC_adj)^(T_end - T0)

equity_value[company, scenario] = NPV - net_debt + non_operating_assets
implied_share_price = equity_value / shares_outstanding
```

Output metric	Definition
EBITDA compression	EBITDA under scenario divided by EBITDA baseline, minus 1, per year
NPV impact	NPV under scenario divided by NPV baseline, minus 1
Implied share-price delta	Compared to current market price
Carbon cost share of EBITDA	Carbon cost in year t divided by baseline EBITDA in year t
Stranded-asset ratio	Sum of stranded asset write-downs divided by book value
Break-even carbon price	Carbon price at which climate-adjusted NPV equals baseline NPV
Scenario divergence	Fan chart of NPV and share price across all four scenarios

7. Known Limitations and Version Scope

The following limitations are explicitly documented. They are framed as the basis for Phase 2 development and for the open questions we would like to validate with domain experts.

- Scope 3 feedback loops into demand are tracked as a separate data layer but are not yet fully priced through the main DCF. They enter the supply-chain disruption path qualitatively.
- Endogenous commodity price formation is out of scope in v0.2. Scenario commodity prices are taken as given inputs from IEA and NGFS; the engine does not model price formation from supply-demand equilibria.
- Portfolio-level correlation between companies in a lending book or fund is a Phase 2 item. The current version produces asset-level and company-level outputs; portfolio aggregation is additive.
- Litigation and reputational risk are qualitative overlay flags only. Pricing these requires actuarial precedent data that is still thin.
- The financial pillar currently relies on disclosed financials and sector-proxy emissions where company disclosure is incomplete. Disclosure gaps are flagged explicitly in outputs rather than silently filled.
- CMIP6 model selection is currently single-model (MRI-AGCM3.2-S) with CMCC-CM2-VHR4 as fallback. Multi-model ensemble averaging is on the roadmap for v0.3 to quantify model uncertainty.

8. Regulatory Alignment

Framework	CRI alignment
TCFD (2017 and 2021)	Physical and transition risk both covered. Scenario analysis across three or more scenarios. Asset-level granularity.
IFRS S2 (2023)	Scope 1 and 2 emissions explicitly modelled. Physical hazard taxonomy consistent with S2 Appendix B. Climate-related financial effects quantified.
CSRD / ESRS E1 (2024)	Scenario-based resilience analysis. Transition plan capex modelling. Value chain emissions tracked.
ECB / EBA Climate Risk Guidelines	Borrower-level physical and transition risk scores. Pillar 3 disclosure support. Horizon to 2050.
NGFS Phase 4	NZE 2050, Delayed Transition, and Current Policies scenarios natively implemented.

9. Data Sources and References

Dataset	Provider	Resolution	Use in CRI
Aqueduct 4.0	World Resources Institute (Kuzma et al. 2023)	Sub-basin approx 10 km	Water stress, riverine flood, coastal flood, drought baselines
CMIP6 Climate Change API	Open-Meteo using MRI-AGCM3.2-S (Mizuta et al. 2017)	0.25 deg approx 25 km	Future temperature and precipitation projections; warming delta
MERRA-2 via NASA POWER	NASA Goddard POWER API	0.5 deg approx 55 km	Observed 2001 to 2020 T2M_MAX, precipitation, humidity baseline

Dataset	Provider	Resolution	Use in CRI
IBTRACS	NOAA IBTrACS v04	Track-level	Cyclone and tropical storm basin and latitude gating
FIRMS Active Fire	NASA FIRMS VIIRS and MODIS	375 m and 500 m	Wildfire fire weather index calibration
Copernicus DEM and SRTM	ESA and NASA	30 m and 90 m	Elevation proxy; slope factor; lapse rate correction
MODIS HLS LULC	NASA MODIS MCD12Q1	500 m	Land cover type; wildfire fuel load; UHI classification
NGFS Phase 4 Scenarios	Network for Greening the Financial System	Global and regional	Carbon price paths; macro scenario alignment
IEA WEO and NZE 2050	International Energy Agency	Sector and global	Commodity demand curves; technology cost trajectories
IPCC AR6 WG1 and WG2	IPCC 2021 and 2022	Assessment	GMST ratios Table 4.5; regional impact functions; sea level
FAO AQUASTAT	Food and Agriculture Organisation	Country and region	Saltwater intrusion baseline; freshwater dependency

Document prepared by ClimRisk | v0.2 | May 2026

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